

CELESTIAL CYCLES

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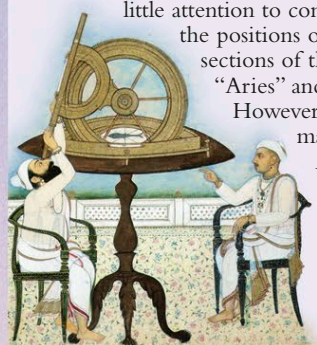
Mapping the sky 348–353

TO AN OBSERVER ON EARTH, celestial events occur within the context of cycles determined by the motions of the Earth, Sun, and Moon. These cycles provide us with some of our basic units for measuring time, such as days and years. They include the apparent daily motions of all celestial objects across the sky, the annual apparent movement of the Sun against the celestial sphere, the seasonal cycle, and the monthly cycle of lunar phases. Other related cycles produce the dramatic but predictable events known as lunar and solar eclipses.

MYTHS AND STORIES

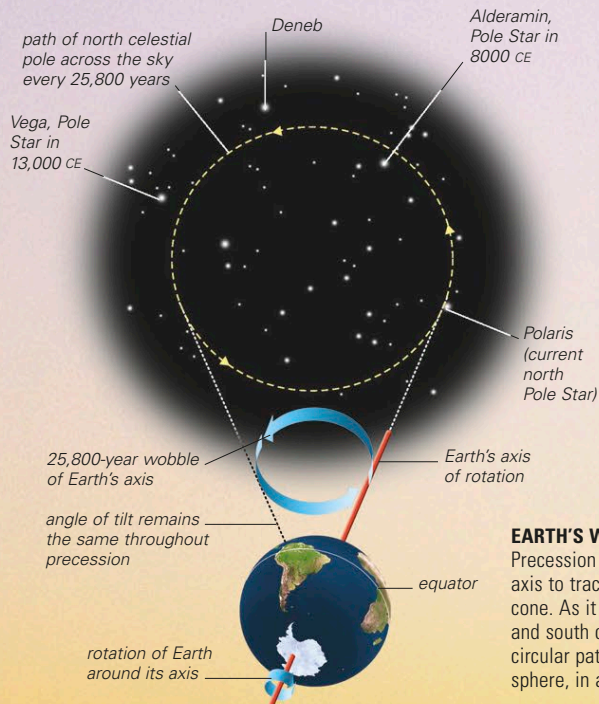
ASTROLOGY AND THE ECLIPTIC

Astrology is the study of the positions and movements of the Sun, Moon, and planets in the sky in the belief that these influence human affairs. At one time, when astronomy was applied mainly to devising calendars, astronomy and astrology were intertwined, but their aims and methods have now diverged. Astrologers pay little attention to constellations, but measure the positions of the Sun and planets in sections of the ecliptic that they call “Aries” and “Taurus,” for example. However, these sections no longer match the constellations of Aries, Taurus, and so on.



STARGAZER

This 17th-century illustration, taken from a treatise written in India on the zodiac, depicts a stargazer using an early form of mounted telescope.



EARTH'S WOBBLE

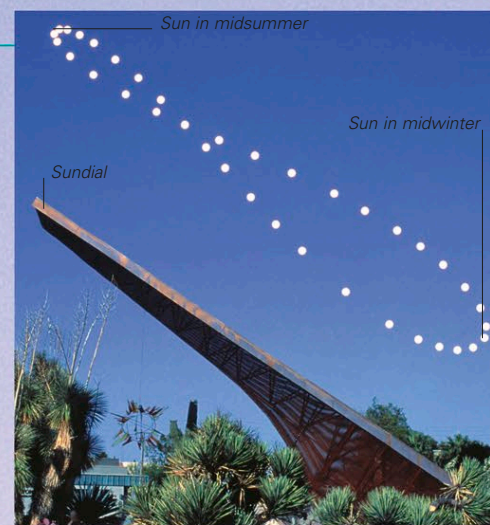
Precession causes Earth's spin axis to trace out the shape of a cone. As it does so, both the north and south celestial poles trace out circular paths on the celestial sphere, in a 25,800-year cycle.

THE SUN'S CELESTIAL PATH

As the Earth travels around the Sun, to an observer on Earth, the Sun seems to trace a path across the celestial sphere known as the ecliptic. Because of the Sun's glare, this movement is not obvious, but the Sun moves a small distance each day against the background of stars. The band of sky extending for 9 degrees (see p.63) on either side of the Sun's path is called the zodiac (see opposite) and incorporates parts or all of 24 constellations (see p.72). Of these, the Sun passes through 13 constellations, of which 12 form the “signs of the zodiac,” well-known to followers of astrology (see panel, left). The Sun spends a variable number of days in each of these 13 constellations. However, the Sun currently passes through each constellation on dates very different from traditional astrological dates. For example, someone born between March 21 and April 19 is said to have the sign Aries, although the Sun currently passes through Aries between April 19 and May 14. This disparity is partly caused by a phenomenon called precession.

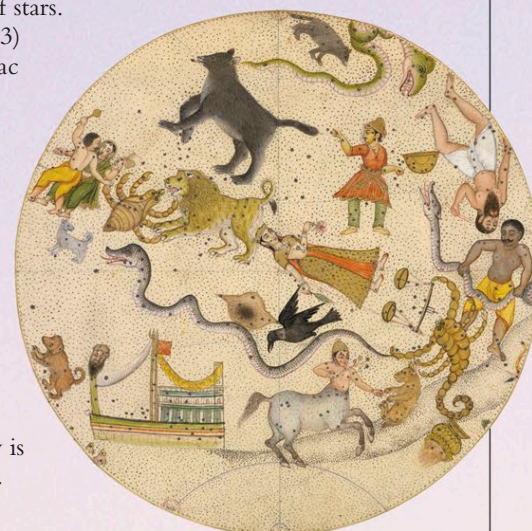
PRECESSION

The Earth's axis of rotation is tilted from the ecliptic plane by 23.5° . The tilt is crucial in causing seasons (see opposite). At present, the axis points at a position on the northern celestial sphere (the north celestial pole) close to the star Polaris, but this will not always be so. Like a spinning top, Earth is executing a slow “wobble,” which alters the direction of its axis over a 25,800-year cycle. The wobble, called precession, is caused by the gravity of the Sun and Moon. It also causes the south celestial pole, the celestial equator, and two other reference points on the celestial sphere, called the equinox points, to change their locations gradually. The coordinates of stars and other “fixed” objects, such as galaxies (see p.63), therefore change, so astronomers must quote them according to a standard “epoch” of around 50 years. The current standard was exactly correct on January 1, 2000.



THE SUN'S ANALEMMA

To produce this image, the Sun was photographed, above a sundial, at the same time of day on 37 occasions throughout 1 year. The vertical change in its position is due to Earth's tilt. The horizontal drift is due to Earth changing its speed on its elliptical orbit around the Sun. The resulting figure-eight pattern is called an analemma.



ISLAMIC ZODIAC

This Islamic depiction of part of the celestial sphere includes several constellations that are also well-known zodiacal “star signs,” such as Scorpius and Leo. The illustration decorates a 19th-century manuscript from India that brought together Islamic, Hindu, and European knowledge of astronomy.

MIDNIGHT SUN

This multiple-exposure photograph (below) shows the path of the Sun around midnight near the summer solstice in Iceland. Since the photograph was taken in polar latitudes, Earth's angle of tilt ensures the Sun does not set.